



SASTRA
ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION
DEEMED TO BE UNIVERSITY
(U/S 3 of the UGC Act, 1956)



THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

MODELING AND ANALYSIS OF IEEE 13 BUS UNBALANCED DISTRIBUTION SYSTEM USING OPENDSS WITH PV INTEGRATION

MINIPROJECT - FINAL REVIEW PRESENTATION

TEAM MEMBERS:

**1.DINESH GOVIND G C
2.AKASH S
3.ADHITYA A V**

GUIDED BY:

**Sr. Asst. Professor
Dr. KARTHIKAIKANNAN D**

ABSTRACT

This work presents the analysis of the IEEE 13-Bus unbalanced distribution system by evaluating the p.u. voltage profile and power losses under different operating conditions. The benchmark system is compared with a modified system incorporating additional loads and capacitor compensation under a 24-hour varying load profile. Further, photovoltaic (PV) integration is analyzed at weak buses such as Bus 675, 680, 611, and 652 to study voltage regulation and loss reduction. The results show that PV integration improves voltage profile and reduces active and reactive power losses by decreasing feeder current flow. Among the analyzed buses, Bus 675 shows the most effective improvement in voltage stability and loss minimization.

OBJECTIVES

- To model and analyze the IEEE 13-Bus unbalanced distribution system in OpenDSS and study voltage profile, power flow, and system losses under varying load conditions.
- To evaluate the impact of additional practical loads and capacitor compensation on feeder performance, voltage stability, and loss reduction.
- To integrate photovoltaic (PV) systems at weak buses such as 675, 611, 652, and 680 and compare their effects on voltage improvement and line losses.
- To identify the optimal PV placement for improving overall system efficiency, reducing feeder losses, and maintaining stable operation in the distribution network.

LITERATURE REVIEW



REFERENCE	SOURCE	INFERENCE
IEEE13 – Bus Analysis in OpenDSS with PV System and Storage – IEEE Xplore	IEEE XPLORE https://ieeexplore.ieee.org/document/129592/	1.Addition of PV reduced voltage drop across the feeder but increased phase imbalance. 2.Large loads shift voltages up or down depending on demand-generation present on the particular bus. 3.Distributed Energy sources provided voltage support but the unbalancing was still present.
Unbalanced Distribution Power Flow with Distributed Generation Sarika Khushalani, Student Member, IEEE, and Noel Schulz, Senior Member, IEEE –RESEARCH GATE	RESEARCH GATE https://www.researchgate.net/publication/224643624_Unbalanced_Distribution_Power_Flow_with_Distributed_Generation	This paper works on matching the calculated data with the specified data of standard test case data (with and without DG integration) and cross-checking results using RDAP-Radial Distribution Analysis Package used for load flow analysis
Impact Assessment of PV Penetration on Unbalanced Distribution network with Dynamic load condition- IEEE Xplore	IEEE XPLORE https://ieeexplore.ieee.org/document/9726552/	Optimal Placement and sizing of PV (100KW at Bus 675) improved positive sequence voltage profiles and reduced line losses compared without PV, while maintaining system parameters within (ANSI) permissible limits.

ADHERENCE TO CODES, STANDARDS & ETHICAL PRACTICES (IEEE 13-BUS SYSTEM)

- **DISTRIBUTION SYSTEM VALUES**

1. This study utilized the IEEE 13-bus radial distribution system, a standard benchmark defined by IEEE PES.
2. The original configuration, load data, and line parameters were taken from standard IEEE resources. This feeder is widely recognized as a standard benchmark system for load flow studies

- <https://cmte.ieee.org/pes-testfeeders/resources/>

- **CAPACITOR METHOD USAGE**

Capacitor placement is performed using reactive power compensation (P-Q method) to improve system power factor and voltage profile. The required shunt capacitor injection is computed using:

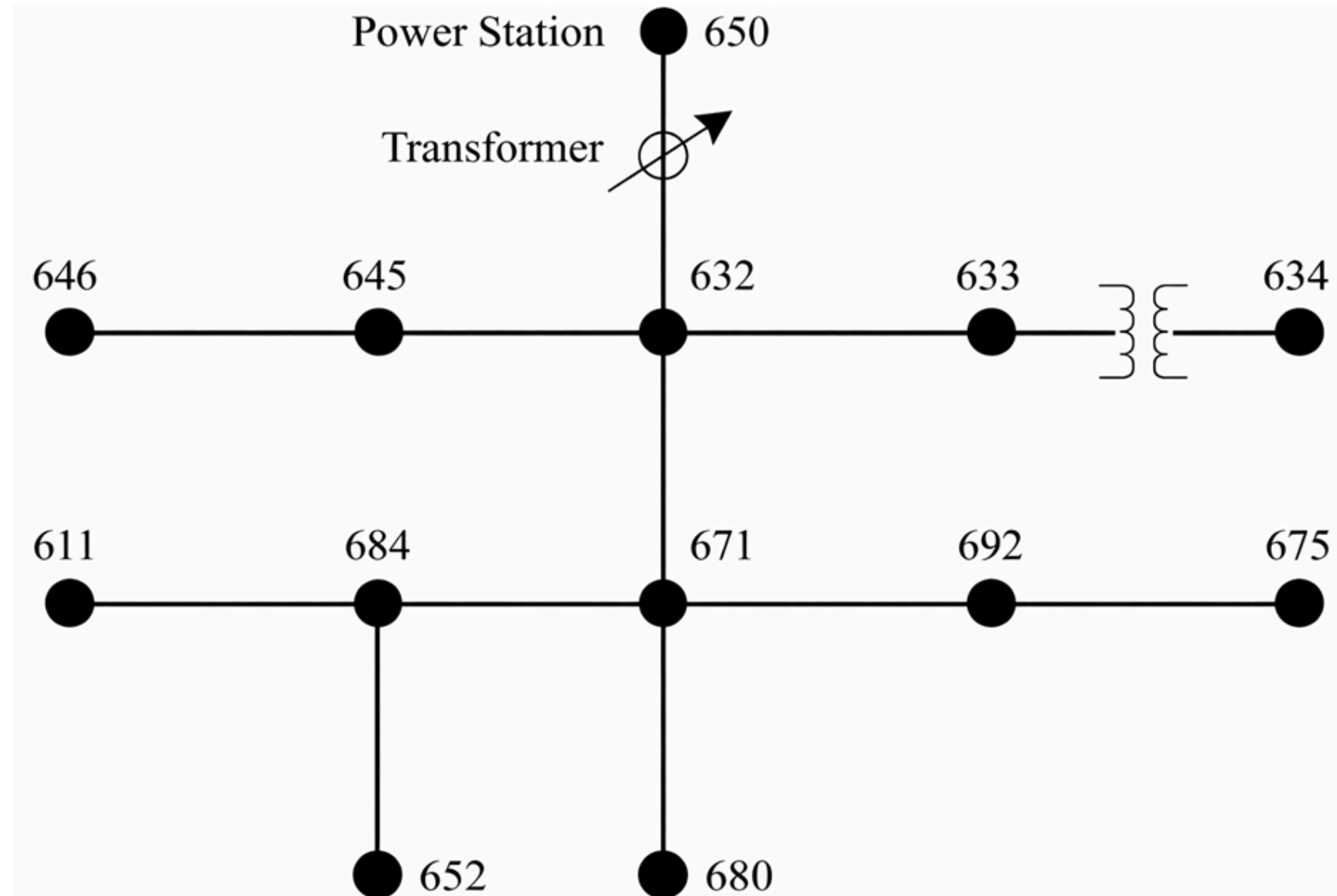
$$Q_{cap} = P (\tan(\phi_1) - \tan(\phi_2))$$

This ensures reduction of reactive power demand from the upstream feeder.

- <https://www.theelectricalguy.in/tutorials/3-steps-to-calculate-required-reactive-power-for-pf-correction/>

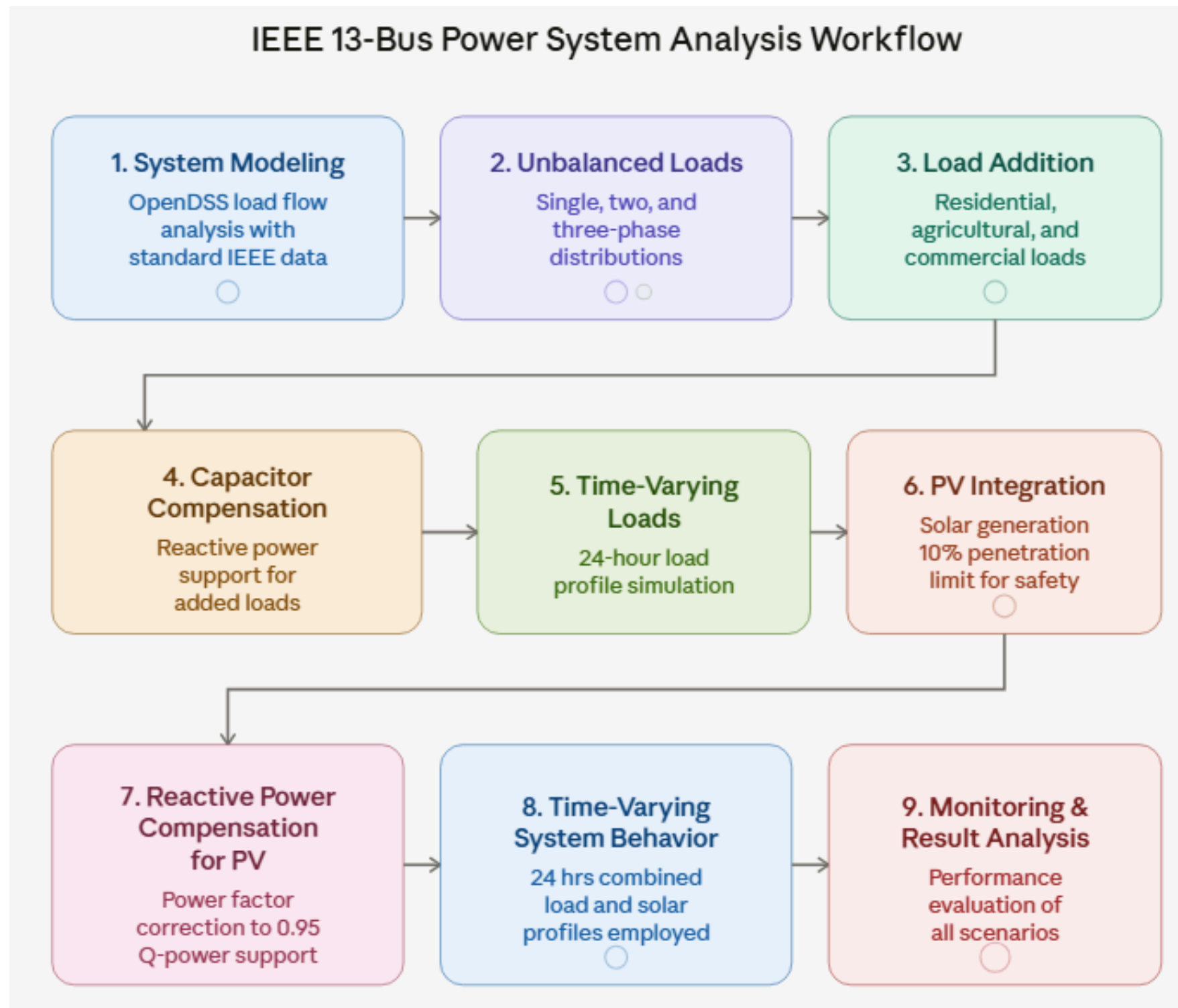
INTRODUCTION

IEEE 13 BUS DISTRIBUTION SYSTEM



- The IEEE 13-bus is an unbalanced radial distribution feeder modeled in OpenDSS. It operates at 4.16 kV and is supplied from a 115 kV source through a 5 MVA step-down transformer between the substation and primary feeder.
- The system includes three-phase, two-phase, and single-phase lines, loads, voltage regulators, and a distribution transformer, representing a practical rural unbalanced network.

METHODOLOGY



- **OpenDSS - system modelling** load flow analysis implementation with standard IEEE data
- **Load Addition** -Single phase ,two phase and three phase loads like Residential, agricultural, and commercial loads added to the existing system to increase the imbalance in loads.
- **Capacitor Compensation** - Reactive power support for added loads.
- **24-hour load profile** employed.
- **PV Integration** - Solar PV with 10% penetration limit for safe penetration and overvoltage protection.
- **Capacitor added** - Reactive Power Compensation for PV for Q-power support
- **24 hrs-load and solar profiles** employed.
- **Comparative analysis** were done for boht condition and **Result were observed**

IMPLEMENTATION

LOADS ADDITION

LOAD ALTERATION AND CAPACITOR COMPENSATION

LOADS	Bus Connected	Phases	Connection	Voltage (kV)	Active Power (kW)	Reactive Power (kVAR)
Irrigation Pump -motor load	671.1	1	Wye	2.4	80	40
Residential Houses-Fans, lights, AC	680.1.2.3	3	Wye	4.16	150	90
Cooling systems(AC)- Schools	684.3	1	Wye	2.4	60	30
Agro Processing Mill	692.1.2.3	3	Wye	4.16	120	70

CAPACITOR COMPENSATION

CAPACITORS	Bus Connected	Phases	Voltage (kV)	Reactive Power (kVAR)
Cap3	671.1	1	2.4	40
Cap4	680	3	4.16	100
Cap5	684.3	1	2.4	30
Cap6	692	3	4.16	80

- As a result more current absorption will be take place.
- To balance the current losses compensation capacitors were added at subsequent places

IMPLEMENTATION



PV INTEGRATION - REASON FOR SELECTION OF BUSES

BUS 675

Bus 675 is electrically weak due to its downstream location and unbalanced single-phase loading, causing higher voltage drop. PV is integrated at this bus for local power support and voltage regulation analysis.

BUS 680

Bus 680 is close to the feeder backbone and is mainly influenced by upstream loading and regulator action. PV placement is used to study feeder power flow characteristics.

Bus 611

Bus 611 is supplied through a single-phase lateral with low load demand and capacitor compensation, resulting in smaller voltage deviations. PV integration is considered under lightly loaded conditions.

BUS 652

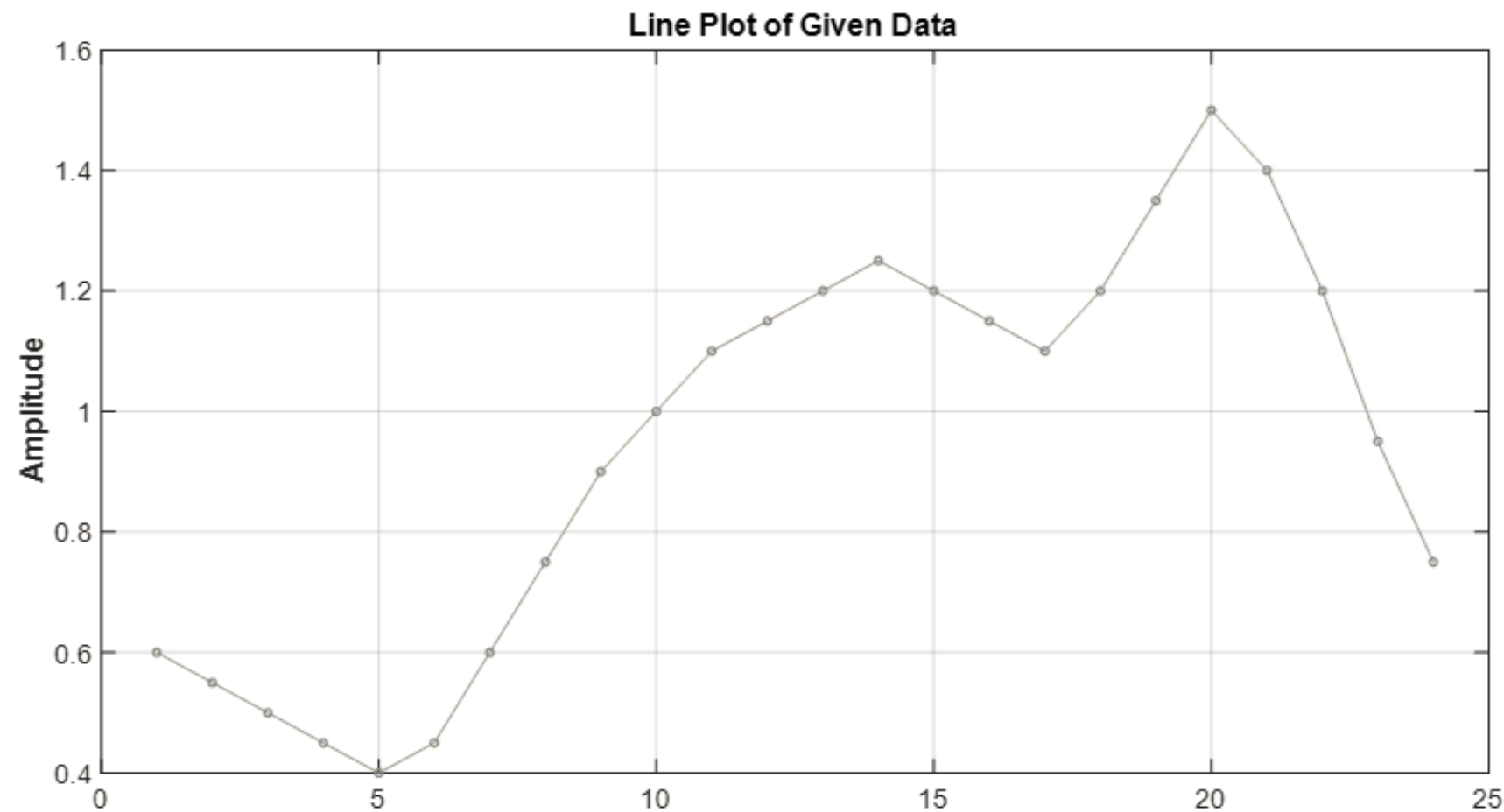
Bus 652 is a weak end bus connected through a high-impedance single-phase lateral, leading to significant voltage drop. PV is integrated to analyze downstream feeder behavior.

IMPLEMENTATION



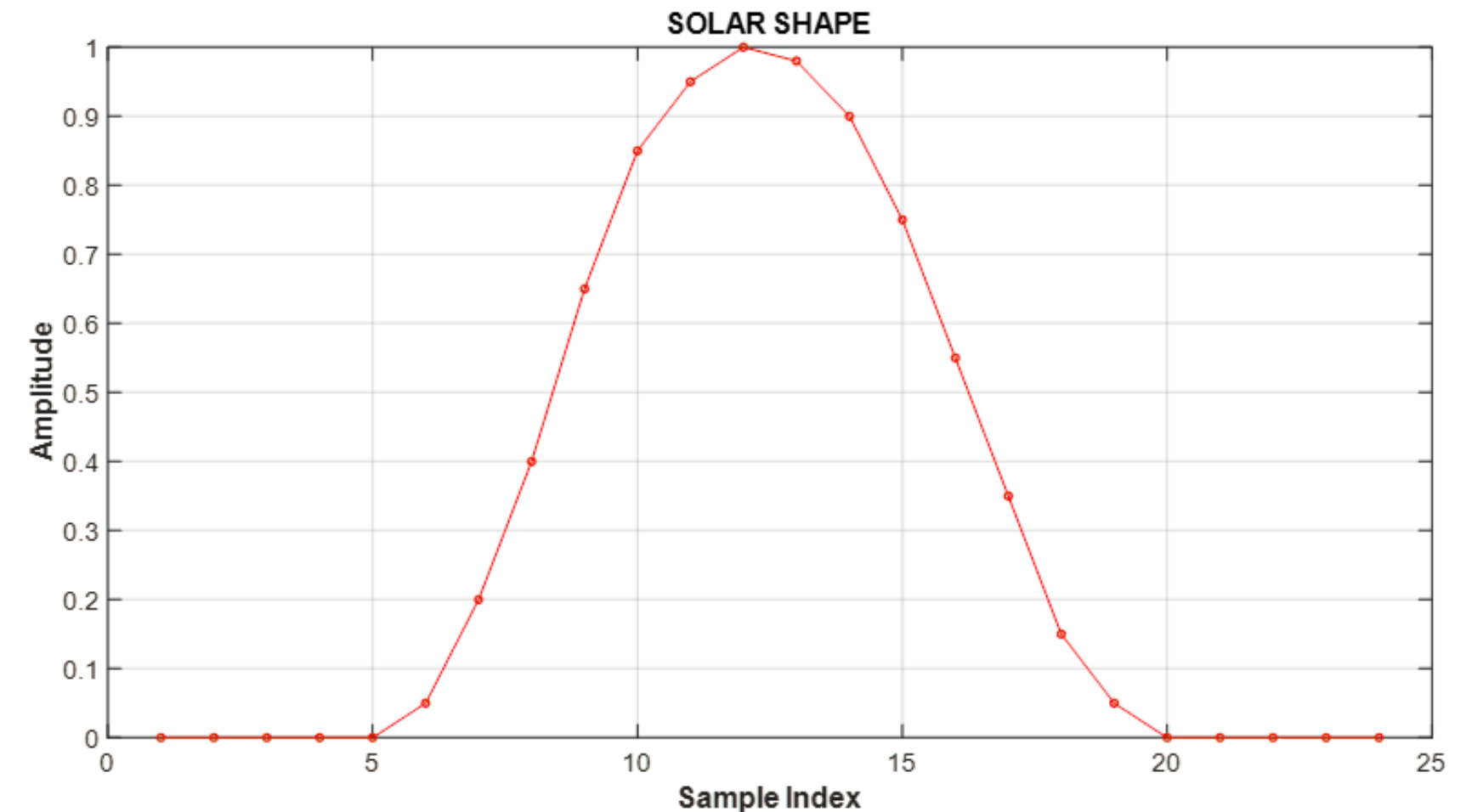
LOAD AND SOLAR SHAPE

LOAD SHAPE VARIATION



- Morning rise: due to lighting, cooking, and water usage
- Daytime dip: people are away (farming/work) → minimal appliance use
- Evening peak: maximum usage (lighting, TV, cooking, household activities)
- Night decline: reduced activity → load decreases

SOLAR SHAPE VARIATION

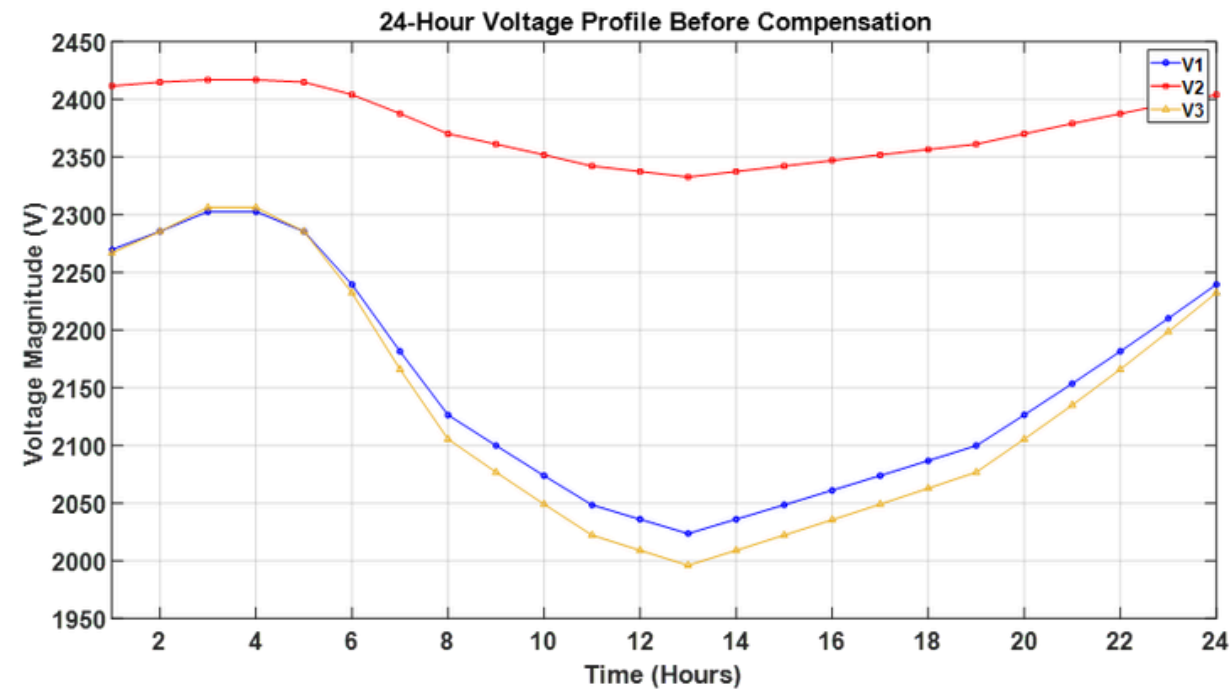


- Zero output (night): no sunlight → no generation (early & late hours)
- Morning rise: gradual increase in solar irradiance after sunrise
- Midday peak: maximum sunlight → highest generation (~1.0 p.u.)
- Afternoon fall: Intensity decreases toward sunset
- Evening to night: output returns to zero

RESULTS

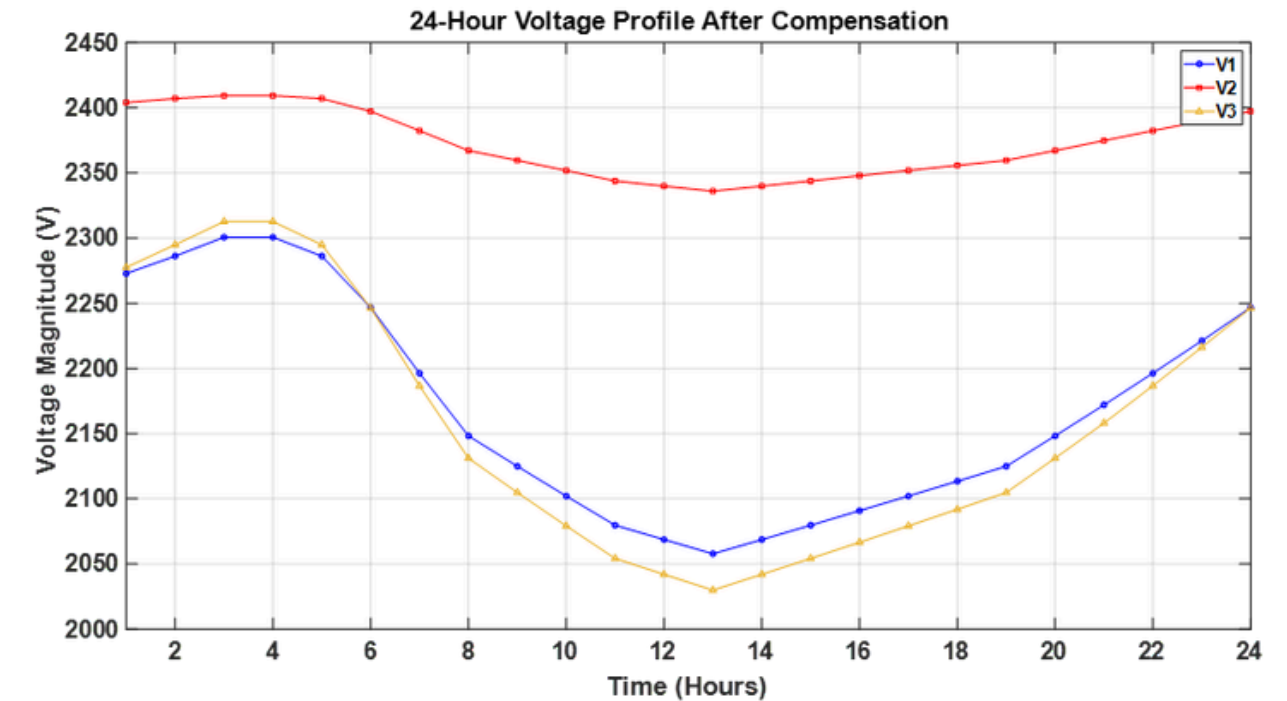
LOAD ADDITION AND CAPACITOR COMPENSATION

VOLTAGE VARIATION (LINE 632-671) ON A DAY



BEFORE ADDITION OF NEW LOADS WITH CAPACITOR COMPENSATION

- **Phase A:** Higher load demand causes considerable voltage drop and poor voltage regulation in Phase A.
- **Phase B:** Phase B shows moderate voltage fluctuation due to feeder loading and system imbalance.
- **Phase C:** Heavy reactive loading leads to severe voltage degradation in Phase C.



AFTER ADDITION OF NEW LOADS WITH CAPACITOR COMPENSATION

- **Phase A:** Capacitor support reduces voltage drop and improves Phase A voltage regulation.
- **Phase B:** Reactive power compensation improves voltage consistency in Phase B.
- **Phase C:** Local capacitor compensation enhances voltage stability and reduces voltage drop in Phase C.

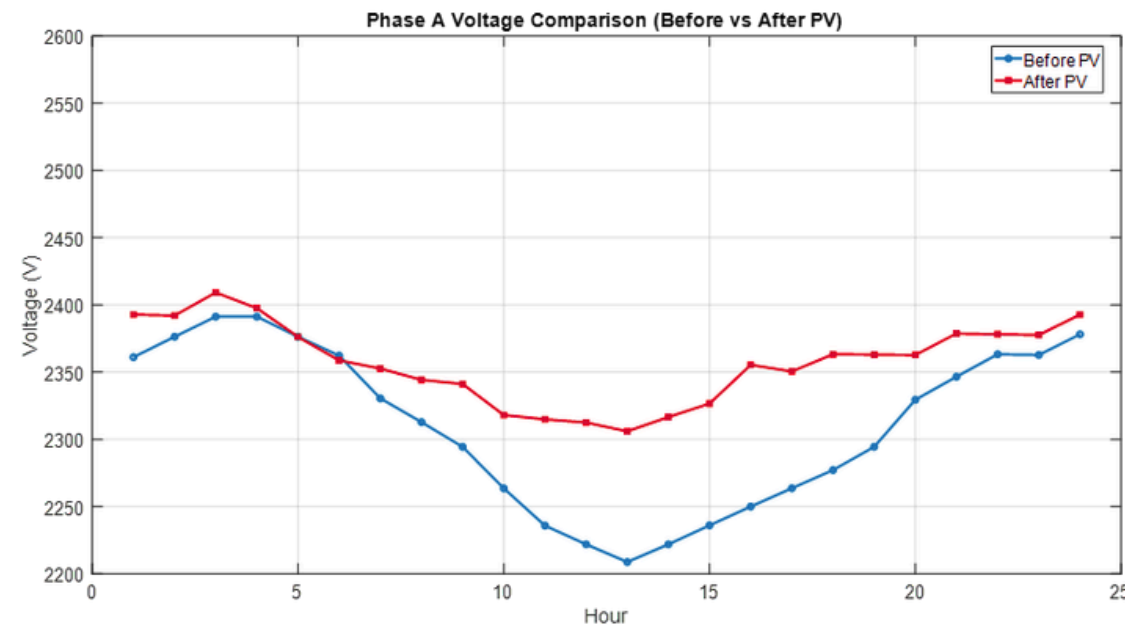
RESULTS

PHASE VOLTAGE IMPACT ON BUS 675

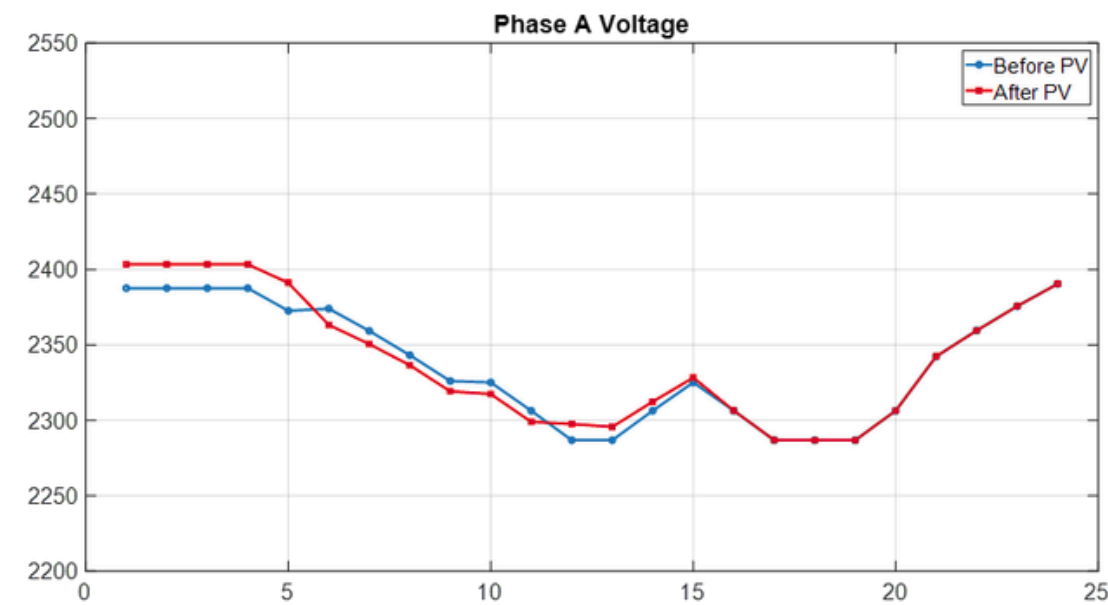


● AFTER PV INJECTION
● BEFORE PV INJECTION

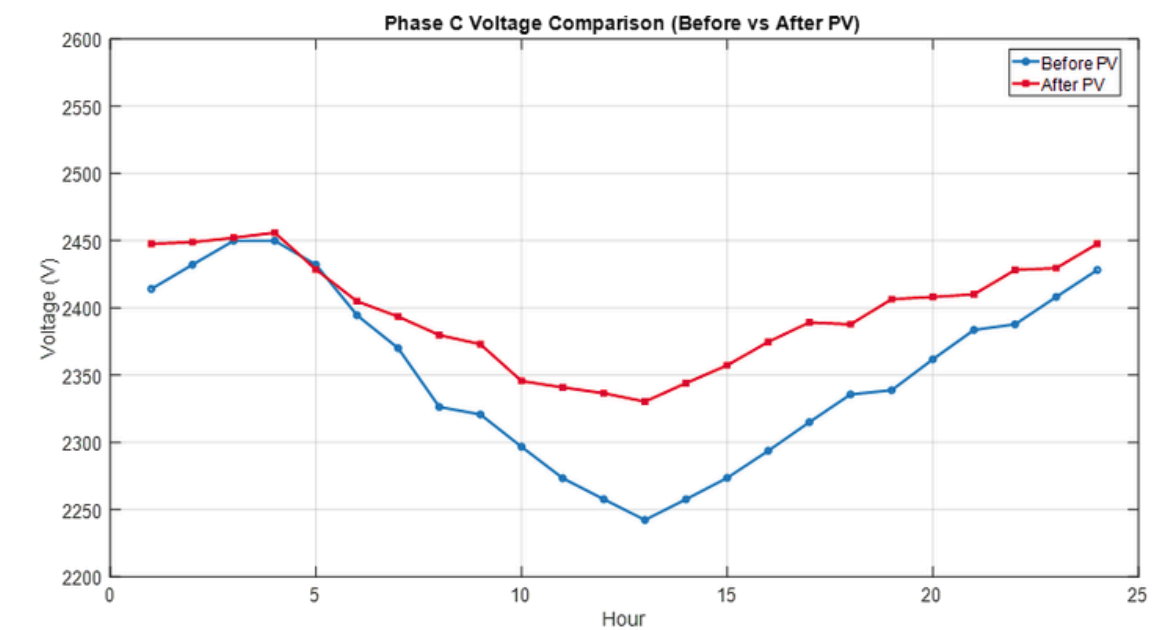
PHASE A



PHASE B



PHASE C



INFERENCE:

BEFORE PV

Voltage drops during peak hours due to high feeder current causing I^2R losses reducing the voltage profile. Since it is a mid-feeder bus, it carries power to many downstream loads, so it is strongly affected by loading.

AFTER PV

Voltage improves clearly and becomes more stable. PV reduces feeder current and gives strong support to the system, leading to maximum loss reduction. During low load, reverse power flow occurs, but it is controlled.

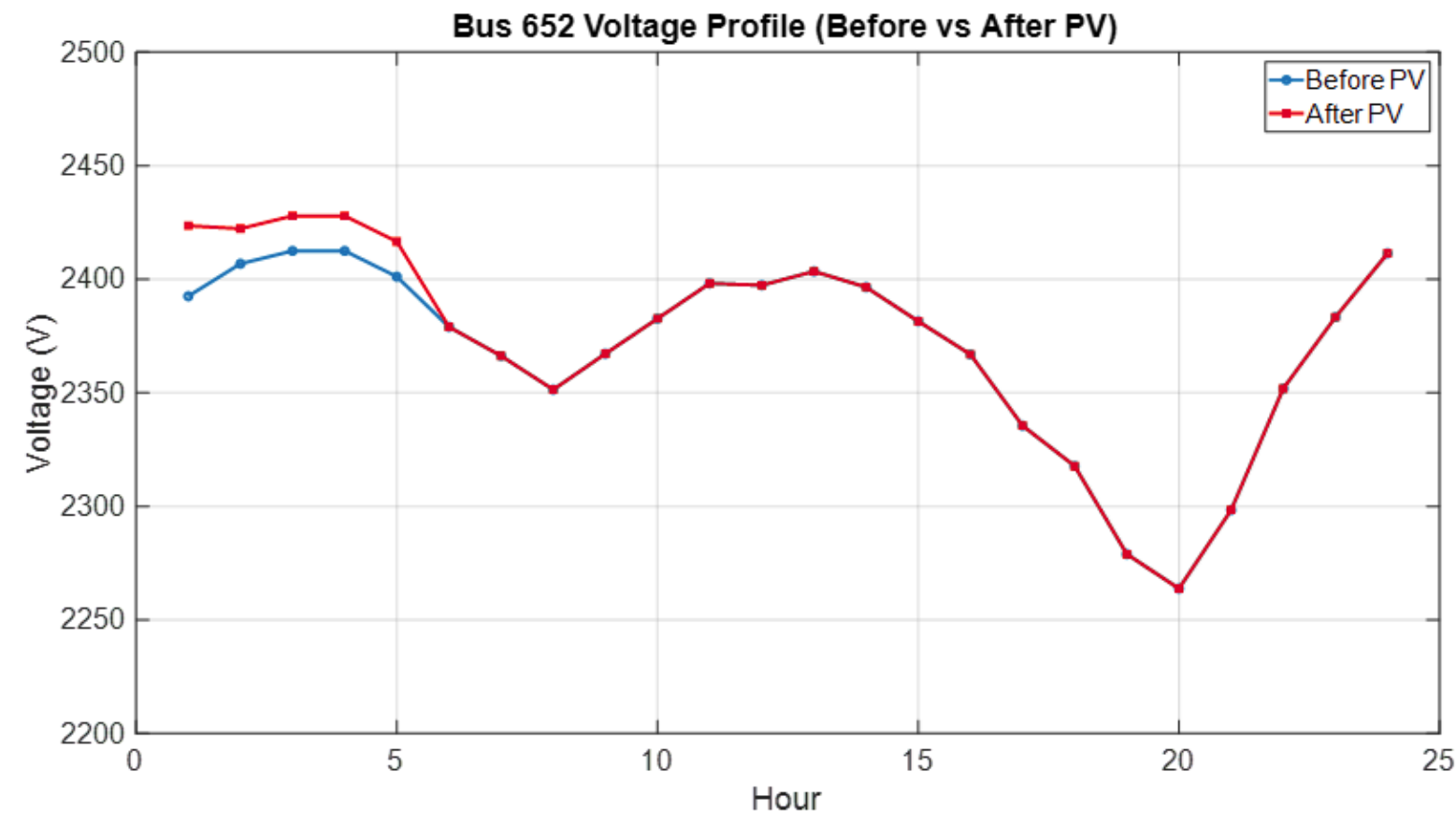
RESULTS

PHASE VOLTAGE IMPACT ON BUS 652



● AFTER PV INJECTION
● BEFORE PV INJECTION

PHASE A



INFERENCE:

BEFORE PV

Voltage is lower during peak load because it is at the end of the feeder, where voltage drop is high due to long distance and high impedance effect.

AFTER PV

At Bus 652, PV impact is very small because the bus is far from the source and the line impedance is high, so voltage improvement is minimal and not clearly visible in the graph

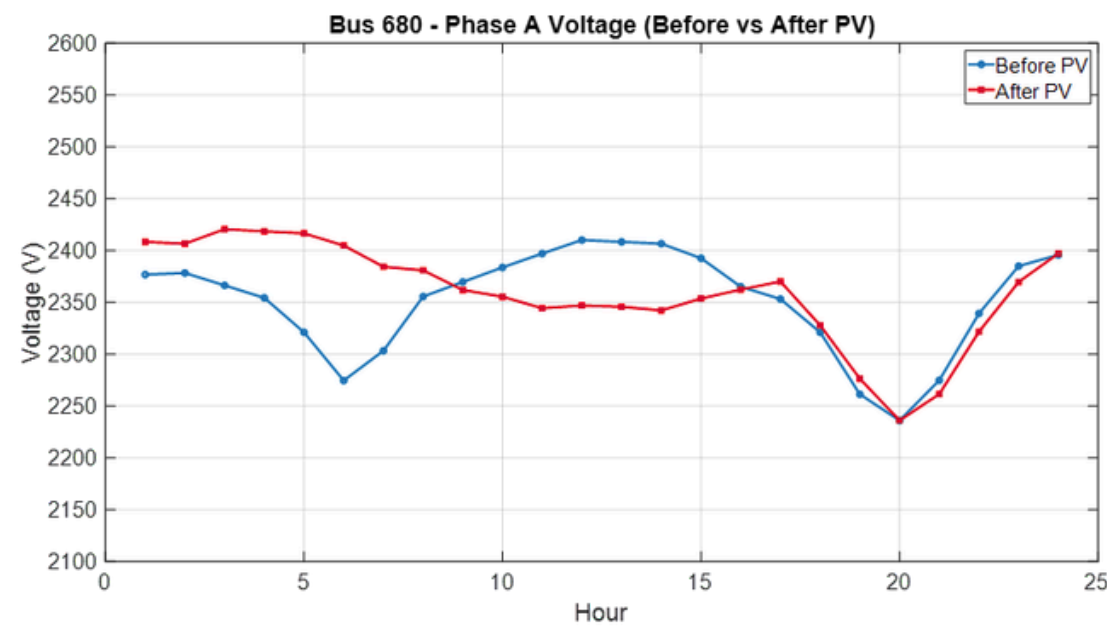
RESULTS

PHASE VOLTAGE IMPACT ON BUS 680

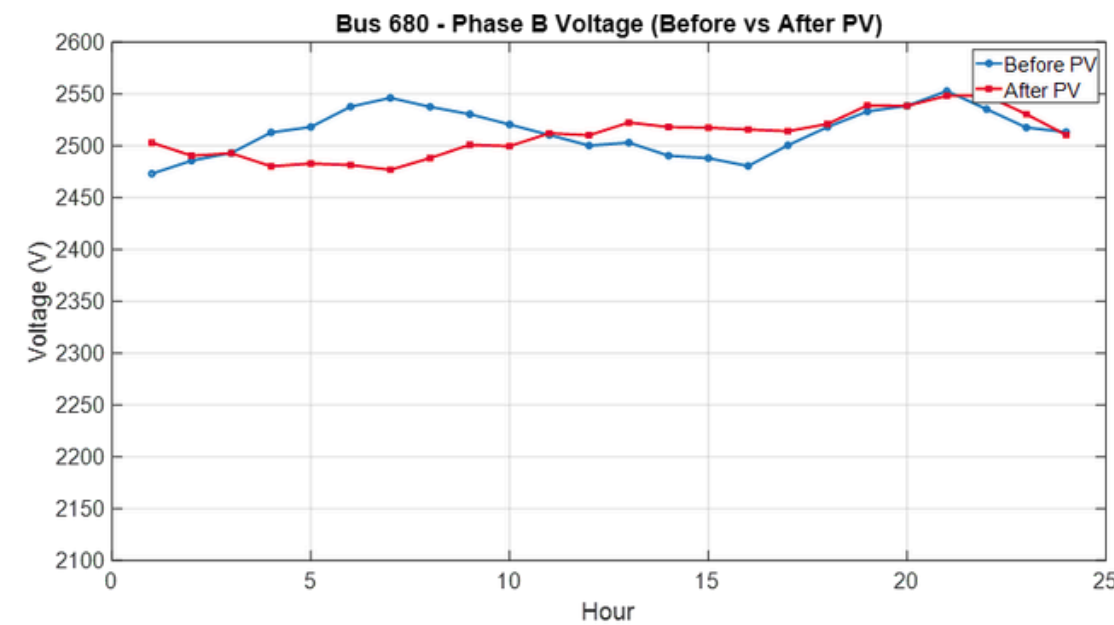


● AFTER PV INJECTION
● BEFORE PV INJECTION

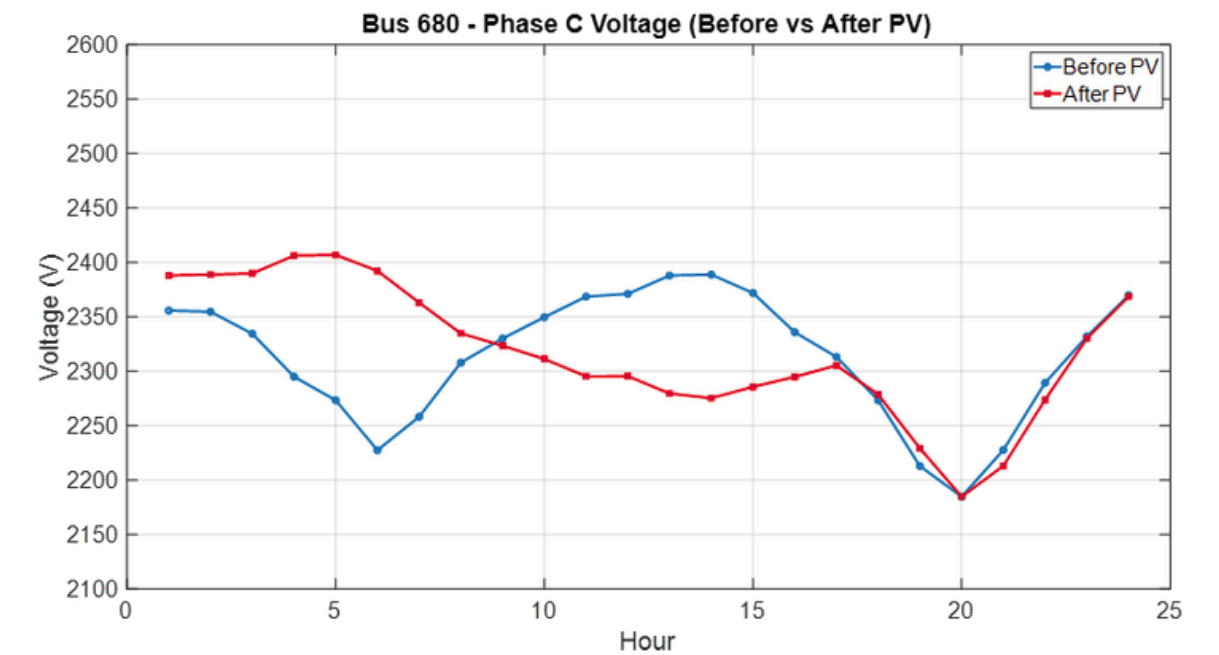
PHASE A



PHASE B



PHASE C



INFERENCE:

BEFORE PV

Voltage drops moderately during peak load due to its location at the feeder end/lateral branch. Impact is less than central buses but still noticeable.

AFTER PV

Voltage improves due to local PV support. However, the improvement is not uniform, and reverse power flow appears at some hours.

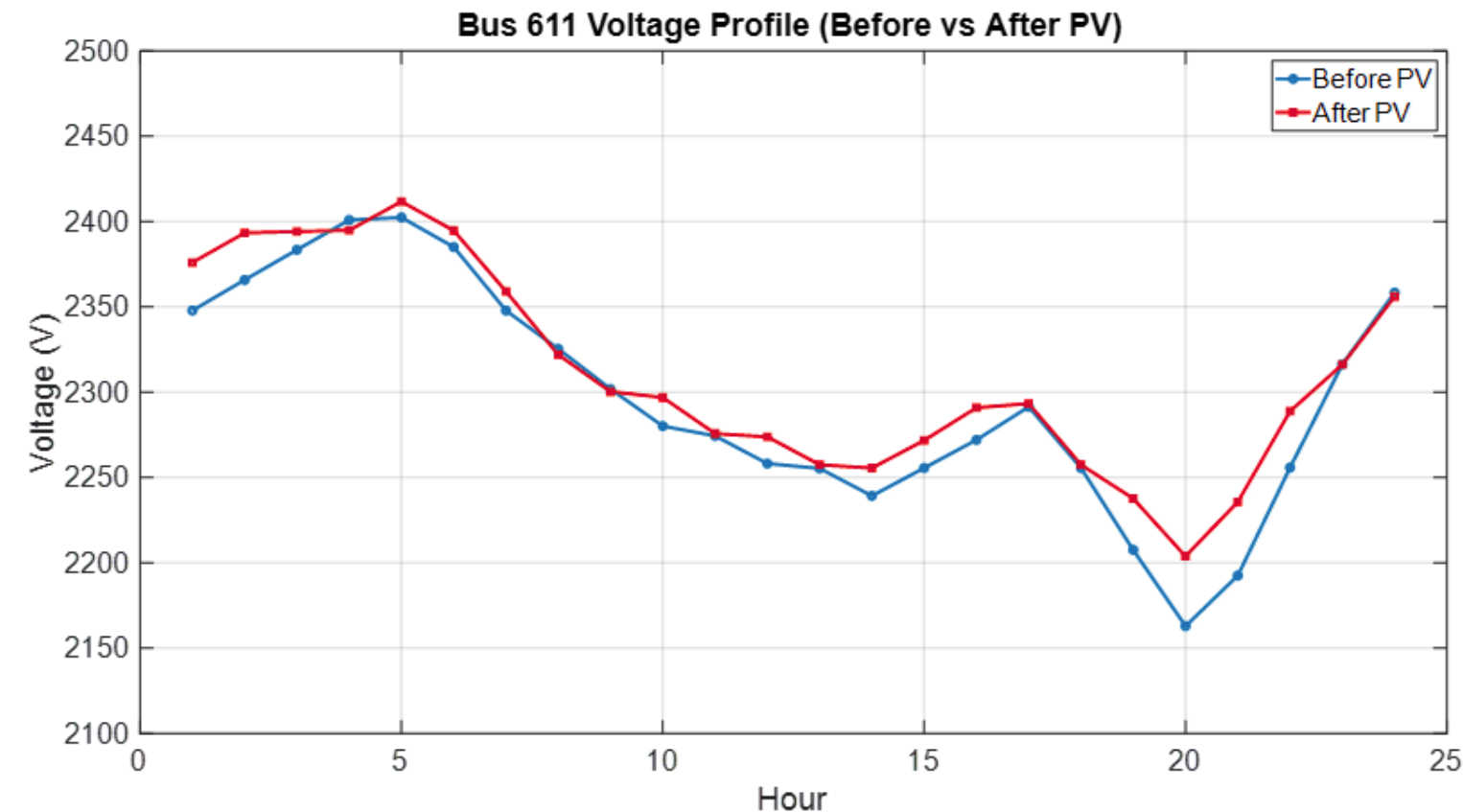
RESULTS

PHASE VOLTAGE IMPACT ON BUS 611



● AFTER PV INJECTION
● BEFORE PV INJECTION

PHASE C



INFERENCE:

BEFORE PV

Voltage is mostly stable because this bus is close to the substation and secondary distribution (Bus 671). Changes depend mainly on total system load.

AFTER PV

Only small improvement is seen because voltage is already strong. Reverse power flow effect is almost negligible here.

CAPACITOR CALCULATION

PROCEDURE



1: Data Collection AND Target PF OF 0.95 IS SET:

P and Q for each phase were collected from the load connected to that bus.
The desired $PF_2 = 0.95$ was set.

2: Initial Reactive Factor Calculation:

The initial **reactive factor** for each phase was calculated using:
 $\tan\phi_1 = Q / P$

3: Capacitor Sizing & Selection:

Capacitor sizes were calculated separately for all three phases.

$$Q_{\text{cap}} = P \times (\tan\phi_1 - \tan\phi_2)$$

Substituting $\tan\phi_2 = 0.328$:

$$Q_{\text{cap}} = P \times (\tan\phi_1 - 0.328)$$

The nearest standard capacitor rating was selected.

CAPACITOR CALCULATION



PROCEDURE

FOR ADDITIONAL LOAD

The target power factor is taken as $PF_{\text{target}} = 0.95$, which gives $\tan\phi_2 = 0.328$.

The capacitor sizing formula used is:

$$Q_c = P(\tan\phi_1 - \tan\phi_2), \text{ where } \tan\phi_1 = Q/P.$$

For Bus 671.1

$$Q_c = P_{671.1} (Q_{671.1} / P_{671.1} - 0.328) = 40 \text{ kVAR.}$$

For Bus 680

$$Q_c = P_{680} (Q_{680} / P_{680} - 0.328) = 100 \text{ kVAR.}$$

For Bus 684.3

$$Q_c = P_{684.3} (Q_{684.3} / P_{684.3} - 0.328) = 30 \text{ kVAR.}$$

For Bus 692

$$Q_c = P_{692} (Q_{692} / P_{692} - 0.328) = 80 \text{ kVAR.}$$

FOR PV Q-SUPPORT

The capacitor ratings are calculated based on power factor correction to 0.95. which gives $\tan\phi = 0.328$

- Bus 675: $P \approx 485 \text{ kW}$, initial $Q \approx 190 \text{ kV AR}$

$$Q_c = P \tan\phi - Q \Rightarrow \text{Capacitor} \approx 310 \text{ kV AR}$$

- Bus 611: $P \approx 170 \text{ kW}$, initial $Q \approx 80 \text{ kV AR}$

$$Q_c \approx 24 \text{ kV AR}$$

PV CALCULATION

PV SIZING USING TOTAL SYSTEM LOAD

TOTAL SYSTEM LOAD:

$$P_{\text{total}} = 3266 \text{ kW}$$

PV is typically sized as a percentage of total load (standard practice: 5% to 20% for distribution systems)

Assume 10% penetration:

$$P_{\text{PV}} = 0.10 \times 3266$$

$$P_{\text{PV}} = 326.6 \text{ kW}$$

PRACTICAL PV RATING SELECTION:

$$P_{\text{PV}} \approx 300 \text{ kW}$$

(rounded to a standard, feasible value)

Technical Reason

- 300 kW is close to 10% of total feeder load → ensures safe penetration
- Avoids oversizing → prevents overvoltage and reverse power issues
- Provides effective loss reduction and voltage improvement

PV CALCULATION PROCEDURE

PV SIZING METHOD USING MINIMUM LOAD MULTIPLIER

Determine the total peak load at the selected bus by summing the loads connected at that bus.

$$P_{\text{peak}} = \sum P_{\text{loads}}$$

Multiply the P_{peak} with the min multiplier of (24 hrs)load shape.

$$P(t) = P_{\text{peak}} \times \text{mult}(t)$$

where $P(t)$ is the load at time t , P_{peak} is the peak load, and $\text{mult}(t)$ is the hourly load multiplier.

Calculate the minimum load condition using the minimum multiplier from the 24-hour load profile.

$$P_{\text{min}} = P_{\text{peak}} \times \text{mult}_{\text{min}}$$

Limit the PV penetration level such that PV generation does not exceed the minimum load demand.

$$P_{\text{PV}} \leq P_{\text{min}}$$

This constraint helps avoid reverse power flow and overvoltage conditions.

Select PV capacity approximately equal to the minimum load value.

$$P_{\text{V_cap}} \approx P_{\text{min}}$$

PV CALCULATION

PV SIZING BASED ON MINIMUM LOAD MULTIPLIER ON A PV BUS

- **LOAD DATA AT BUS 675**

The total load at Bus 675 is obtained from three single-phase loads:

$$P_{675a} = 485 \text{ kW}$$

$$P_{675b} = 68 \text{ kW}$$

$$P_{675c} = 290 \text{ kW}$$

Total Peak Load:

$$P_{\text{peak}} = 485 + 68 + 290$$

$$P_{\text{peak}} = 843 \text{ kW}$$

- **24-HOUR LOAD VARIATION MODEL**

The time-varying load is modeled using the given multiplier profile:

$$P(t) = P_{\text{peak}} \times \text{mult}(t)$$

Where:

$$P(t) = \text{load at time } t$$

$$\text{mult}(t) = \text{hourly load multiplier}$$

- **MINIMUM LOAD CALCULATION**

From the given 24-hour profile:

$$\text{Minimum multiplier} = 0.40$$

Therefore:

$$P_{\text{min}} = P_{\text{peak}} \times 0.40$$

$$P_{\text{min}} = 843 \times 0.40$$

$$P_{\text{min}} = 337.2 \text{ kW}$$

- **PV SIZING CONSTRAINT (MINIMUM LOAD METHOD)**

The PV penetration is limited by:

$$P_{\text{PV}} \leq P_{\text{min}}$$

To avoid:

1. Reverse power flow
2. Overvoltage at the bus

- **MAXIMUM PV CAPACITY AT BUS 675**

$$P_{\text{PV,max}} \approx 337.2 \text{ kW}$$

Thus the PV sizing is chosen as 300 kW close to the calculated value

THANK YOU